

SEMESTER EXAMINATION-2022
CLASS - PAPER CODE: BET-C610
PAPER TITLE: Embedded Systems

Time: 3 hour

Max. Marks: 70

Note: Question Paper is divided into two sections: **A and B**. Attempt both the sections as per given instructions.

SECTION-A (SHORT ANSWER TYPE QUESTIONS)

Instructions: Answer any five questions in about 150 words each. Each question carries six marks.
(5 X 6 = 30 Marks)

Question-1: Discuss the basic differentiate between timers and counters in case of 8051 microcontroller.

Question-2: What is an embedded system? What are the components of embedded system?

Question-3: Discuss Operating system Architecture approach. Which approach of OS designing is widely used these days explain.

Question-4: Discuss the role of Optimized program in embedded system Design.

Question-5: Draw and compare Von-Neumann and Harvard architecture approaches of processor designing.

Question-6: What are the points to be considered while connecting power supply rails with embedded system?

Question-7: Discuss Addressing modes in 8051 microcontroller .

Question-8: State the difference between FSM and FSMD models.

Question-9: Discuss the goals of RTOS in short?

Question-10: Write a program in C language to interface 8 LED's with microcontroller 8051 on any port of microcontroller 8051 / 89c51.

SECTION-B (LONG ANSWER TYPE QUESTIONS)

Instructions: Answer any FOUR questions in detail. Each question carries 10 marks.
(4 X 10 = 40 Marks)

Question-11: Write a program (a) to make P1 an input port, (b) to get a byte of hex data in the range of 00- FFH from P1 and convert it to decimal. Save the digits in R7, R6 and R5, where the least significant digit is in R7 .

Question-12: A switch is connected to pin P1.7 (8051 microcontroller). Write a program to check the status of the switch and make the following decision.

- a) if SW=0, send '0' to P2
- b) if SW=1 send '1' to P2

Question-13: Explain with diagram of embedded system development life cycle.

Question-14: Write a program to interface LCD (Liquid Crystal Display unit) with microcontroller 8051.

Question-15: Analyze the following two types of loop and determine the number of iterations to be performed and number of steps per iterations.

(a) Loop 1
int sum=0;
For (int j=0; j<N; j++)
Sum=sum+j;

(b) Loop 2
Int sum=0;
For (int j=0; j<100; j++)
Sum=sum+j;

Question-16: Write a Program to find the GCD of two numbers, how can you optimize it.

Question-17: Explain the block diagram of special purpose microprocessor. Also discuss the differences between RISC and CISC registers.

Question-18: Define NRE Cost in short. The life time of a product is 58 weeks. If the product is delayed by 5 weeks, determine the percentage revenue loss? Determine the per product cost if NRE cost is Rs. 500000 and unit cost is Rs. 8000 and company produces 6000 product units of that product

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